JEPPESEN®
A BOEING COMPANY

GUIDED FLIGHT DISCOVERY

PRIVATE PILOT SYLLABUS

Introduction

This syllabus utilizes the building-block method of teaching in which each instructional item is presented on the basis of previously learned knowledge and skill. It guides students and instructors through a lesson sequence in which new material builds on what a student has already mastered.

The Private Pilot Course contains separate ground and flight training courses. Therefore, the course may be conducted as a combined ground and flight training program or be divided into separate components. When using the integrated sequence shown in the time allocation tables, aeronautical knowledge that is pertinent to a flight lesson is taught just before that flight.

GROUND TRAINING

Ground school is an integral part of pilot certification courses and the Ground Training Syllabus meets FAR Part 141 requirements for ground training. When coordinating the ground school with flight training, each ground lesson is conducted at the point indicated in the Allocation Tables beginning on page xvii. Ground training Stages I and II are completed during Stage I of the flight training syllabus. Ground Stage III, and the End-of-Course Final Exams "A" and "B" are completed during Stage II of flight training. This sequence enables the student to complete the aeronautical knowledge segments of the syllabus before the final stage of flight training, and encourages the student to take the FAA Private Pilot Airman Knowledge Test promptly. Appendix B identifies the specific ground lessons in which knowledge areas required by Part 141 and Part 61 are covered.

In a classroom environment, ground lessons should normally be presented in numerical order. However, to provide some flexibility for adapting to individual student needs and to the training situation, the order of the lessons may be altered with approval of the chief flight instructor. Any deviation should not disturb the course continuity or objectives. Each lesson may be presented in one classroom session, or it may be divided into two or more sessions, as necessary.

GROUND TRAINING COMPONENTS

The Guided Flight Discovery Pilot Training System provides the necessary components to train for a private pilot certificate. Students and instructors should review the following list as a guide to assemble and effectively use the course materials.

Private Pilot Syllabus—the outline of the Private Pilot course. The syllabus provides a basic framework for training in a logical sequence and assigns appropriate study material prior to each lesson.

Private Pilot Textbook/e-Book—the primary source for initial study and review. Textbook chapters are references for each ground lesson. The textbook contains complete and concise explanations of the fundamental concepts and ideas that every private pilot needs to know. The subjects are organized in a logical manner to build upon previously introduced topics. Subjects are often expanded upon through the use of Discovery Insets, which are strategically placed throughout the chapters. The Summary Checklists, Key Terms, and Questions are designed to help students review and prepare for both the knowledge and practical tests.

Private Pilot Maneuvers Manual—provides illustrated step-by-step instructions for all required flight maneuvers. As study assignments prior to specific flight lessons, maneuvers lessons save training time by preparing students to perform the maneuver and reducing the time required for preflight briefing of maneuvers. Maneuvers are numbered for ease of reference, are grouped into categories based on similar operational characteristics, and presented in the order in which they are typically introduced during training.

Private Pilot Online Course—provides academic content in ground lessons with exams and interactive maneuvers lessons in a complete ground school. Online lessons are listed as references for ground lessons in this syllabus. Lessons use a combination of audio, video and graphics to clearly explain each topic. Maneuvers lessons provide step-by-step guidance with in-cockpit video showing what the pilot sees when performing each maneuver. A Learning Management System (LMS) tracks completions and test results specific to each question to assist in identifying student strengths and weaknesses.

Jeppesen FAR/AIM—includes the current Federal Aviation Regulations (FARs) and the Aeronautical Information Manual (AIM) in one publication available in printed form or as an e-Book. The FAR/AIM includes FAR Parts 1, 3, 11, 43, 48, 61, 67, 68, 71, 73, 91, 97, 103, 105, 107, 110, 119, 135, 136, 137, 141, 142, NTSB 830, and TSRs 1552 and 1562. The AIM is a complete reproduction of the FAA publication with full-color graphics and the Pilot/Controller Glossary. The AIM contains basic flight information and ATC procedures to operate effectively in the U.S. National Airspace System.

Private Pilot Exams—provide essential testing components. Required by FAR Part 141, students take a stage exam after each stage of ground training and End-of-Course Final Exams at the completion of ground training. Prior to solo flight, students must take the Presolo Exam required by FAR Part 61.

Private Pilot Airman Knowledge Test Guide—helps students understand the learning objectives for the test questions so that they can take the FAA Private Pilot Airman Knowledge Test with confidence. The test guide contains sample knowledge test questions, with correct answers, explanations, and study references. Explanations of why the other choices are wrong are included where appropriate.

LESSON EXAMS

Each ground lesson has a brief exam at the end. Students may complete the exam online in the Jeppesen Learning Center or complete the questions in the appropriate textbook chapter. As specified in the ground lesson completion standards, the exams must be scored and the student must discuss any incorrect responses with the instructor to ensure student understanding prior to beginning the next ground lesson. When a lesson is complete, the instructor assigns the next textbook chapter and section(s) or online lesson(s) for out-of-class study.

STAGE EXAMS

At the end of each stage, the student is required to successfully complete the stage exam outlined in the syllabus before beginning the next ground training stage. The stage exam evaluates the student's understanding of the knowledge areas within a stage. Successful completion of each stage exam and a review of each incorrect response is required before the student progresses to the next stage.

END-OF-COURSE FINAL EXAMS

When all of the ground lesson assignments are complete, the student should take the End-of-Course Final Exams "A" and "B" assigned in Stage III. Following the exams, the instructor should assign the student appropriate subject areas for review. After a thorough review, the student should complete the FAA Private Pilot Airman Knowledge Test without delay.

USING AN ATD FOR GROUND TRAINING

The syllabus provides for use of an ATD for ground training. An ATD is extremely effective for introducing procedures without the distraction of having to fly the airplane. If properly integrated into the ground training program, the ATD will enhance systems knowledge and procedural understanding before the student trains in the airplane.

In addition to skill enhancement, the introduction of maneuvers and procedures by instrument reference in the ATD has other advantages for both student and instructor. These include fewer distractions, more versatility in lesson presentation, repositioning, freeze functions, emergency training, and the ability to control the environment of the training session and allow the student to concentrate on the areas the instructor wants to emphasize. By following the recommended sequence of the syllabus, the student will gain maximum benefit from the integration of academic training, introduction of new maneuvers and procedures in the ATD, and subsequent practice in the airplane. The following ground lessons are particularly suited to the use of an ATD.

Ground Lesson 1 — Discovering Aviation	(1 Hour)
Ground Lesson 2 — Airplane Systems	(1 Hour)
Ground Lesson 5 — Communication and Flight Information	(1 Hour)
Ground Lesson 12 — Navigation	(1 Hour)
Ground Lesson 14 — Flying Cross-Country	(1 Hour)

If the box for ATD utilization is checked in the Preface, then the ATD becomes part of the ground training segments for the approved course, and its use is required. If the box for ATD utilization is left blank, the ATD is not part of the approved course, and its use is not required. Not checking the ATD box does not preclude a ground instructor from using an ATD. An ATD may be used in ground training just like any other classroom instructional aid.

FLIGHT TRAINING

The Flight Training Syllabus also is divided into three stages, each providing an important segment of pilot training. Tables in Appendix B specify the stages and specific flight lessons in which each flight training task required by Part 141, Part 61, and the FAA *Private Pilot — Airplane Airman Certification Standards (ACS)* is introduced, reviewed, and evaluated.

Each stage builds on previous learning and should be completed in sequence. However, to provide flexibility for adapting to individual student needs and the training environment, the syllabus lesson sequence may be altered with approval of the chief flight instructor. Any deviation should not disturb the course continuity or objectives. The Preface also includes two options for adjusting the lesson sequence to allow the first solo to occur in Stage II of flight training. If either box is checked, then the optional sequence is required for flight training.

STAGE I

In Stage I, the student develops the knowledge, skill, and habits necessary for safe solo flight. The basic maneuvers are introduced, practiced, and reviewed. In addition, the student practices airport operations, normal and crosswind takeoffs and landings, emergency procedures, and ground reference maneuvers. FAR 61.87 specifies the solo requirements for student pilots including the aeronautical knowledge that must be tested by the Presolo Exam and the maneuvers and procedures required for presolo training.

NOTE: The student must complete the Presolo Exam and Briefing prior to the first solo flight, regardless of whether that flight occurs at the end of Stage I or in Stage II.

STAGE II

This stage introduces short- and soft-field takeoffs, climbs, approaches, and landings; VOR, GPS, and ADF navigation (based on aircraft equipment); and night flying. The maneuvers introduced during this stage incorporate the skills developed during Stage I, and provide important skills necessary for the cross-country operations later in this stage.

The cross-country portion of this stage provides the information, knowledge, and skills that enable the student to begin cross-country operations. With the knowledge acquired during Stage II, the student should be able to safely conduct solo cross-country flights. Proficiency in advanced maneuvers and cross-country procedures will be evaluated during the Stage II Check.

STAGE III

The flights of Stage III are designed to provide the student with the proficiency required for the private pilot practical test. These flights are devoted to gaining experience and confidence in cross-country operations and reviewing all maneuvers within the syllabus to attain maximum pilot proficiency. Student proficiency and knowledge will be assessed by the chief instructor, assistant chief instructor, or check instructor during the Stage III Check. The student can pursue further review and instruction as necessary, in preparation for the End-of-Course Flight Check.

PREFLIGHT DISCUSSION

Prior to each dual and solo flight, the instructor should provide the student with a thorough briefing of the tasks to be covered during the lesson. The instructor should define unfamiliar terms, explain the maneuvers and objectives of each lesson, and discuss single-pilot resource management (SRM) concepts related to each lesson. The Preflight Discussion should be tailored to the specific flight, the local environment, and especially to the needs of the student.

AIRPLANE PRACTICE

Each flight should begin with a review of previously-learned maneuvers before any new maneuvers are introduced. To ensure that students efficiently utilize solo flight training lessons, the instructor should train the student in the maneuvers to be performed during the flight and discuss what is expected to be accomplished during that lesson.

FSS, FTD, OR ATD

The syllabus allows for instruction in a flight simulation training device (FSTD), which is defined as a full flight simulator (FSS) or flight training device (FTD). Training in an FSS that meets the requirements of 141.41(a) may be credited for a maximum of 20 percent of the total flight training hour requirements (20%× 35 hours = 7.0 hours). Training in an FTD that meets the requirements of 141.41(b) may be credited for a maximum of 15 percent of the total flight training hour requirements (15% × 35 hours = 5.25 hours). Training in an FSS or FTD, if used in combination, may be credited for a maximum of 20 percent of the total flight training hour requirements However, credit for training in an FTD cannot exceed 15 percent.

The regulations do not specifically address the use of an ATD to meet private pilot flight training requirements. However, according to FAR 61.4(c), the FAA may approve a device other than an FSS or FTD for specific purposes. An ATD may be used for flight training credit if the school obtains an FAA letter of authorization (LOA). AC 61-136A — *FAA Approval of Aviation Training Devices and Their Use for Training and Experience* provides guidelines for schools to obtain an LOA that approves an ATD for meeting up to 15% of the total flight training requirements.

POSTFLIGHT DEBRIEFING

The Postflight Debriefing is as important as the Preflight Discussion. The student should perform a self-critique of maneuvers/procedures and SRM skills. This *learner-centered grading* is especially helpful in developing decision-making skills. If the student is having trouble mastering a skill, both the student and instructor should plan for improving performance. An effective Postflight Briefing increases retention and helps the student prepare for the next lesson. As a guide, a minimum of 1/2 hour per flight is recommended for the Preflight Discussion and Postflight Briefing combined.

STUDENT STAGE CHECKS

Stage checks measure the student's accomplishments during each stage of training. The conduct of each stage check is the responsibility of the chief instructor. However, the chief instructor may delegate authority for conducting stage checks and the End-of-Course Flight Check to the assistant chief instructor or a designated check instructor. This procedure provides close supervision of training and provides another opinion on the student's progress. The stage check also helps the chief instructor check the effectiveness of the instructors.

An examination of the building-block theory of learning will show that it is extremely important for progress and proficiency to be satisfactory before the student enters a new stage of training. Therefore, the next stage should not begin until the student successfully completes the stage check. Failure to follow this progression might defeat the purpose of the stage check and degrade the overall effectiveness of the course.

PILOT BRIEFINGS

The following three pilot briefings are assigned in the flight syllabus.

1. Presolo Exam and Briefing
2. Solo Cross-Country Briefing
3. Private Pilot Practical Test Briefing

Pilot briefings are contained in Appendix A of this syllabus. Each briefing consists of a series of questions to prepare the student for the knowledge and tasks required in subsequent solo lessons and the practical test. The student should review the appropriate briefing questions in advance and research the answers to gain optimum benefit from the briefing.

The briefings should be conducted as private tutoring sessions to test each student's comprehension. Hold the briefings in a comfortable classroom or office environment, and schedule ample time. Discuss every question thoroughly to ensure the student understands the key points. Correct placement of the briefings is indicated in the Allocation Tables

The Presolo Exam and Briefing is unique. As specified in FAR 61.87, the student must demonstrate satisfactory knowledge of the required subject areas by completing a knowledge exam. This exam is to be administered and graded by the instructor who endorses the student pilot certificate for solo flight and must include questions on applicable portions of FAR Parts 61 and 91. In addition, instructors should modify the exam as necessary to make it appropriate for the aircraft to be flown and the local flying environment.

PART 61 OPERATION

The *Private Pilot Syllabus* is designed to meet all the requirements of Part 141, Appendix B. It can be adapted to meet the flight time (airplane, single-engine) requirements of FAR 61.109. The basic difference between the flight time requirements of Part 141 and Part 61 is that under Part 61, the student must have at least 40 hours of flight time that includes at least 20 hours of flight instruction from an authorized instructor and 10 hours of solo flight training (in specified areas of operation). The flight time requirements of Part 141 are nearly the same, except total flight time is only 35 hours. Adapting this syllabus to Part 61 training requires only a slight modification of individual flight lesson times.

The ground training requirements under Part 61 specify that an applicant for a knowledge test must have a logbook endorsement from an authorized instructor who conducted the training or reviewed the applicant's home study course. The endorsement must indicate satisfactory completion of the ground instruction or home study course required for the certificate or rating being sought. A home study course for the purposes of Part 61 is a course of study in those aeronautical knowledge areas specified in FAR 61.105, and organized by a pilot school, publisher, flight or ground instructor, or by the student. The Jeppesen Private Pilot Course satisfies this requirement. As a practical consideration, students seeking pilot certification under Part 61 should receive some formal ground training, either in the classroom or from an authorized flight or ground instructor.

CREDIT FOR PREVIOUS TRAINING

According to FAR 141.77, when a student transfers from one FAA-approved school to another approved school, course credits obtained in the previous course of training may be credited for 50 percent of the curriculum requirements by the receiving school. However, the receiving school must determine the amount of credit to be allowed based upon a proficiency test or knowledge test, or both, conducted by the receiving school. A student who enrolls in a course of training may receive credit for 25 percent of the curriculum requirements for knowledge and experience gained in a non-Part 141 flight school, and the credit must be based upon a proficiency test or knowledge test, or both, conducted by the receiving school. The amount of credit for previous training allowed, whether received from an FAA-approved school or other source, is determined by the receiving school. In addition, the previous provider of the training must certify the kind and amount of training given, and the result of each stage check and end-of-course test, if applicable.

Course Overview

The Private Pilot Course is designed to coordinate the academic study assignments and flight training required to operate in an increasingly complex aviation environment. New subject matter is introduced and evaluated during the ground lessons using the following methods, as chosen by your school.

1. In-depth textbook assignments with study questions:
   - ◊ *Private Pilot* textbook/e-Book
   - ◊ *Private Pilot Maneuvers* manual/e-Book
2. The Private Pilot online course in the Jeppesen Learning Center
3. Instructor/student discussions
4. Stage exams and end-of-course exams

For best results, students should complete ground lessons just prior to the respective flight lessons, as outlined in the syllabus. However, it is also acceptable to present lessons in a formal ground school before introducing the student to the airplane. If significant time has elapsed between the ground lesson and the associated flight, the instructor should conduct a short review of essential material. Generally, flight lessons should not be conducted until the student has completed the prerequisite ground lessons.

In selected flight lessons, the abbreviation "VR" indicates that students should maintain aircraft control by using outside visual references. "IR" indicates that they should use instrument references. No indication of either "VR" or "IR," means that normal private pilot maneuvers or procedures should be conducted by outside visual references.

THE PRIVATE PILOT CERTIFICATION COURSE AIRPLANE SINGLE-ENGINE LAND

COURSE OBJECTIVES

The student will obtain the knowledge, skill, and aeronautical experience necessary to meet the requirements for a private pilot certificate with an airplane category rating and a single-engine land class rating.

COURSE COMPLETION STANDARDS

The student must demonstrate through knowledge exams and flight checks, and show through appropriate records that he/she meets the knowledge, skill, and experience requirements necessary to obtain a private pilot certificate with an airplane category rating and a single-engine land class rating.

STUDENT INFORMATION

COURSE ENROLLMENT

There are no prerequisites for initial enrollment in the ground portion of the Private Pilot course.

REQUIREMENTS FOR SOLO FLIGHT

Before you may fly solo, you must hold a recreational, sport, or student pilot certificate and at least a current third-class medical certificate. You must be at least 16 years of age to obtain a student pilot certificate and be able to read, speak, write, and understand the English language. Remember that solo flight operations require specific training, successful completion of a presolo knowledge exam, and endorsements from your flight instructor.

REQUIREMENTS FOR GRADUATION

To graduate, you must be at least 17 years of age, be able to read, speak, write, and understand the English language, meet the time requirements listed in the Allocation Tables, and satisfactorily complete the training outlined in this syllabus. When you meet the minimum requirements of Part 141, Appendix B, you are eligible for graduation.

LESSON DESCRIPTION AND STAGES OF TRAINING

This syllabus describes each lesson, including the objectives, references, topics, and completion standards. The stage objectives and standards are described at the beginning of each stage within the syllabus.

FLIGHT CHECKS AND GROUND EXAMS

The syllabus incorporates stage checks and the End-of-Course Flight Check as required by Part 141, Appendix B. The chief instructor is responsible for ensuring that each student accomplishes the required stage checks and the End-of-Course Flight Check in accordance with the school's approved training course. However, the chief instructor may delegate authority for stage checks and the End-of-Course Flight Check to the assistant chief or check instructor. You also must complete the academic knowledge tests—stage exams, pilot briefings, and End-of-Course Final Exams—that are described within the syllabus.

Ground Training Overview

Completion of this course is based solely upon compliance with the minimum requirements of FAR Part 141. The accompanying tables with times shown in hours are provided mainly for guidance in achieving regulatory compliance.

PRIVATE PILOT CERTIFICATION COURSE
AIRPLANE SINGLE-ENGINE LAND

	GROUND TRAINING					
	Maneuvers Class Discussion and Online Training	**ATD**	**Ground Lessons**	**Pilot Briefings**	**Stage/ Final Exams**	**Exam Debriefings**
GROUND STAGE I	3.0	3.0	10.0		1.0	As Required
GROUND STAGE II	3.0		6.0	2.0	1.0	As Required
GROUND STAGE III	3.0	2.0	8.0	2.0	4.0	1.0
TOTALS	9.0	5.0	24.0	4.0	6.0	1.0

NOTE:
1. The first column shows the recommended *Private Pilot Maneuvers* class discussion and/or online training time.
2. The second column shows maximum ATD time when an ATD is approved for the course.
3. The third column shows the minimum recommended training time for ground lessons, which may include class discussion or online lessons. Times shown in columns 1 and 2 may be credited toward the total time shown in column 3 as follows:
   - Up to 9 hours of *Private Pilot Maneuvers* class discussion and/or online training.
   - Up to 5 hours of ATD training.

   To recieve credit for ATD training time, the associated course approval must be obtained. (See Preface.)

Allocation Tables

LESSON TIME ALLOCATION													
Ground Training							Flight Training						
							Dual					Solo	
Maneuvers Class Discussion and Online Training	ATD	Ground Lessons	Pilot Briefings	Stage/Final Exams	Exam Debriefings		Day Local	Day Cross-Country	Night Local	Night Cross-Country	Instrument	Day Local	Cross-Country
						GROUND STAGE I, II AND FLIGHT STAGE I							
	1.0	2.0				GL 1 – Discovering Aviation							
	1.0	2.0				GL 2 – Airplane Systems							
						FL 1 – Ground Operations / Basic Maneuvers	.5						
		2.0				GL 3 – Aerodynamic Principles							
1.0						FL 2 – Ground Operations / Basic Maneuvers	1.0						
		2.0				GL 4 – The Flight Environment							
1.0						FL 3 – Slow Flight / Stalls / Basic Instrument	1.0				(.2)		
	1.0	2.0				GL 5 – Communication and Flight Information							
1.0						FL 4 – Emergency Procedures / Steep Turns	1.0				(.2)		
				1.0	As Req.	GL 6 – Stage I Exam							
1.0						FL 5 – Ground Reference Maneuvers	1.0				(.2)		
		2.0				GL 7 – Meteorology for Pilots							
1.0						FL 6 – Takeoffs, Landings, and Go-Arounds	1.0						
		2.0				GL 8 – Federal Aviation Regulations							
1.0						FL 7 – Review	1.0				(.2)		
		2.0				GL 9 – Interpreting Weather Data							
			2.0		As Req.	Presolo Exam and Briefing							
						FL 8 – Review	1.0				(.2)		
						FL 9 – First Solo	.5					.5	
				1.0	As Req.	GL 10 – Stage II Exam							
						FL 10 – Stage I Check	1.0						
6.0	3.0	16.0	2.0	2.0	As Req.	Ground Stage Totals	9.0 (8.5)				(1.0)	.5 (0)	

NOTE:
1. The first column shows the recommended *Private Pilot Maneuvers* discussion and/or online training time.
2. The second column shows maximum ATD time when an ATD is approved for the course.
3. The third column shows the minimum recommended training time for ground lessons, which may include class discussion or online lessons. Times shown in columns 1 and 2 may be credited toward the total time shown in column 3 for the Private Pilot Course as follows:
   - Up to 9 hours of *Private Pilot Maneuvers* class discussion and/or online training.
   - Up to 5 hours of ATD training.

   To recieve credit for ATD training time, the associated course approval must be obtained. (See Preface.)
4. Check the applicable boxes in the Preface to select the following course options. Course approval must be obtained to:
   - Receive credit for ATD training time.
   - Complete the Stage I Check prior to the first solo.
   - Complete Flight Lessons 11, 14, and 15 prior to the first solo.
5. The numbers in parentheses for Dual and Solo Day Local flight times represent total stage times if the Stage 1 check precedes the first solo.

LESSON TIME ALLOCATION

Ground Training							Flight Training						
							Dual					Solo	
Maneuvers Class Discussion and Online Training	ATD	Ground Lessons	Pilot Briefings	Stage/Final Exams	Exam Debriefings		Day Local	Day Cross-Country	Night Local	Night Cross-Country	Instrument	Day Local	Cross-Country
						Ground Stage III and Flight Stage II							
		2.0				**GL 11 – Airplane Performance**							
1.0						**FL 11 – Short/Soft-Field Takeoffs and Landings**	1.0						
	1.0	2.0				**GL 12 – Navigation**							
						FL 12 – Solo Local						1.0	
		2.0				**GL 13 – Human Factor Principles**							
						FL 13 – Solo Local						1.0	
	1.0	2.0				**GL 14 – Flying Cross Country**							
1.0						**FL 14 – Navigation**	1.0				(.5)		
				1.0	As Req.	**GL 15 – Stage III Exam**							
						FL 15 – Navigation Review	1.0				(.5)		
						FL 16 – Night Local			1.0				
1.0						**FL 17 – Dual Cross-Country**		2.0			(.5)		
						FL 18 – Night Cross-Country				2.0	(.5)		
			2.0			**Solo Cross-Country Briefing**							
						FL 19 – Solo Cross-Country							2.5
				3.0	1.0	**GL 16 & 17 – Final Exams A & B**							
						FL 20 – Stage II Check	1.0						
3.0	2.0	8.0	2.0	4.0	1.0	**Stage Totals**	4.0 (4.5)	2.0	1.0	2.0	(2.0)	2.0 (2.5)	2.5

NOTE:
1. The first column shows the recommended *Private Pilot Maneuvers* discussion and/or online training time.
2. The second column shows maximum ATD time when an ATD is approved for the course.
3. The third column shows the minimum recommended training time for ground lessons, which may include class discussion or online lessons. Times shown in columns 1 and 2 may be credited toward the total time shown in column 3 for the Private Pilot Course as follows:
   - Up to 9 hours of *Private Pilot Maneuvers* class discussion and/or online training.
   - Up to 5 hours of ATD training.

   To recieve credit for ATD training time, the associated course approval must be obtained. (See Preface.)
4. Check the applicable boxes in the Preface to select the following course options. Course approval must be obtained to:
   - Receive credit for ATD training time.
   - Complete the Stage I Check prior to the first solo.
   - Complete Flight Lessons 11, 14, and 15 prior to the first solo.
5. The numbers in parentheses for Dual and Solo Day Local flight times represent total stage times if the Stage 1 check precedes the first solo.

LESSON TIME ALLOCATION

Ground Training							Flight Training						
							Dual					Solo	
Maneuvers Class Discussion and Online Training	ATD	Ground Lessons	Pilot Briefings	Stage/Final Exams	Exam Debriefings		Day Local	Day Cross-Country	Night Local	Night Cross-Country	Instrument	Day Local	Cross-Country
FLIGHT STAGE III													
						FL 21 – Solo Cross-Country							2.0
						FL 22 – Solo Cross-Country							4.0
						FL 23 – Practical Test Preparation	2.0				As Req.		
						FL 24 – Practical Test Preparation	2.0				As Req.		
						FL 25 – Stage III Check	1.0						
			As Req.			Private Pilot Practical Test Briefing							
						FL 26 – End-of-Course Flight Check	1.0				As Req.		
						Stage Totals	6.0						6.0
9.0	5.0	24.0	4.0	6.0	1.0	Private Pilot Course – Overall Totals	19.0	2.0	1.0	2.0	(3.0)	2.5	8.5

The individual times shown on the accompanying Lesson Time Allocation tables are for instructor/student guidance only; they are not mandatory for each ground lesson, flight lesson, or stage of training. At the conclusion of this course, the student must meet the minimum requirements of FAR Part 141, Appendix B, for each catagory in order to graduate. Preflight and postflight briefing times are not specified, but a minimum of .5 hours for each dual and solo flight is suggested. The times for Pilot Briefings, although assigned and completed along with selected flight lessons, are considered part of ground training.

Private Pilot Ground Training Syllabus

GROUND TRAINING COURSE OBJECTIVES

The student will obtain the aeronautical knowledge required by FAR Part 141 and Part 61 for private pilot certification and meet the prerequisites in Part 61 for the FAA Private Pilot Airman Knowledge Test.

GROUND TRAINING COURSE COMPLETION STANDARDS

Through knowledge exams and records, the student must demonstrate the aeronautical knowledge required by FAR Part 141 and Part 61 for private pilot certification. The student must also demonstrate the knowledge necessary to pass the FAA Private Pilot Airman Knowledge Test and show that the prerequisites specified in Part 61 have been met.

Stage I

STAGE OBJECTIVES

During this stage, the student is introduced to pilot training, aviation opportunities, and human factors in aviation, and explores airplane systems and aerodynamic principles. The student also learns about the safety of flight, operating at airports, interpreting aeronautical charts, and airspace requirements. In addition, the student gains knowledge about ATC services, radio procedures and how to locate and use sources of flight information.

STAGE COMPLETION STANDARDS

The student must pass the Stage I Exam with a minimum score of 80 percent, and review each incorrect response with the instructor to ensure complete understanding before starting Stage II.

GROUND LESSON 1

REFERENCES

NOTE: Students should study the listed references prior to beginning the Ground Lesson 1 instructional session.

Private Pilot Textbook/e-Book
Chapter 1 — Discovering Aviation (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 1 — Discovering Aviation (GL 1, 2, 3)

OBJECTIVES

- Recognize the essential components of the school's pilot training program.

Part 141/61 Aeronautical Knowledge

- Applicable Federal Aviation Regulations for private pilot privileges, limitations, and flight operations:
  - ◊ Identify the medical and currency requirements for piloting an airplane.
  - ◊ Recognize the requirements to act as pilot in command of different types of aircraft.
- Aeronautical decision making and judgment:
  - ◊ Identify the concepts that apply to single-pilot resource management.
  - ◊ Explain how to perform a self-assessment prior to flight and briefings during flight operations.
  - ◊ Recognize physiological factors that affect your performance during flight.

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- ❑ Aviation Organizations
- ❑ Category and Class Ratings
- ❑ Additional Pilot Certificates
- ❑ Aviation Careers

SECTION C — INTRODUCTION TO HUMAN FACTORS
GL 3 ONLINE — INTRODUCTION TO HUMAN FACTORS

- ❑ Single-Pilot Resource Management
  - ◊ Aeronautical Decision Making and Judgment
  - ◊ Risk Management
  - ◊ Task Management
  - ◊ Situational Awareness
  - ◊ Controlled Flight into Terrain (CFIT) Awareness
  - ◊ Automation Management
- ❑ Aviation Physiology
  - ◊ Pressure Effects
  - ◊ Motion Sickness
  - ◊ Fatigue and Noise
  - ◊ Alcohol, Drugs, and Performance

COMPLETION STANDARDS:

- Demonstrate understanding of policies and procedures that apply to the school's pilot training program.
- Demonstrate understanding of pilot training programs, opportunities in aviation, and human factors during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 1A, 1B, and 1C; or online exams for GL 2 and 3. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 2.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 2 — Airplane Systems (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 2 — Airplane Systems (GL 4, 5, 6)

GROUND LESSON 2

REFERENCES

Private Pilot Textbook/e-Book
Chapter 2 — **Airplane** Systems (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 2 — Airplane Systems (GL 4, 5, 6)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Applicable Federal Aviation Regulations (FARs) for private pilot privileges, limitations, and flight operations:
  - ◊ Identify the inspections and aircraft logbook documentation that are required for airworthiness.
  - ◊ Identify the equipment required for VFR flight under FAR 91.205 and the procedures to fly with inoperative equipment.
- Principles of powerplants and aircraft systems:
  - ◊ Identify airplane components.
  - ◊ Explain how aircraft engines and related systems operate.
  - ◊ Describe flight instrument functions and operating characteristics, including errors and common malfunctions.

CONTENT

SECTION A — AIRPLANES
GL 4 ONLINE — AIRPLANES

❑ The Fuselage
❑ The Wing
❑ The Empennage
❑ Trim Devices
❑ Landing Gear
❑ The Powerplant
❑ Pilot's Operating Handbook (POH)
❑ Airworthiness Requirements

SECTION B — THE POWERPLANT AND RELATED SYSTEMS
GL 5 ONLINE — THE POWERPLANT AND RELATED SYSTEMS

❑ Reciprocating Engine Operation
❑ Induction Systems—Carburetor and Fuel Injection
❑ Supercharging and Turbocharging
❑ The Ignition System
❑ Abnormal Combustion
❑ Fuel Systems
❑ Refueling
❑ Oil Systems

- ❑ Cooling Systems
- ❑ The Exhaust System
- ❑ Propellers—Fixed-Pitch and Constant-Speed
- ❑ Propeller Hazards
- ❑ Full Authority Digital Engine Control (FADEC)
- ❑ Electrical Systems

SECTION C — FLIGHT INSTRUMENTS
GL 6 ONLINE — FLIGHT INSTRUMENTS

- ❑ Pitot-Static Instruments
- ❑ Airspeed Indicator
- ❑ Altimeter
- ❑ Vertical Speed Indicator
- ❑ Gyroscopic Instruments
- ❑ Attitude Indicator
- ❑ Heading Indicator
- ❑ Magnetic Compass
- ❑ Integrated Flight Displays

COMPLETION STANDARDS

- Demonstrate understanding of airplane components and systems, the powerplant and related systems, and flight instruments during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 2A, 2B, and 2C or online exams for GL 4, 5, and 6. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 3.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 3 — Aerodynamic Principles (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 3 — Aerodynamic Principles (GL 7, 8, 9)

GROUND LESSON 3

REFERENCES

Private Pilot Textbook/e-Book
Chapter 3 — Aerodynamic Principles (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 3 — Aerodynamic Principles (GL 7, 8, 9)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Principles of aerodynamics:
  - ◊ Describe the four forces of flight.
  - ◊ Explain the aerodynamic principles and design characteristics that apply to airplane stability and maneuverability.
- Stall awareness, spin entry, spins, and spin recovery techniques:
  - ◊ Recognize the stall and spin characteristics related to training airplanes.
  - ◊ Explain how to recognize and recover from stalls and spins.

CONTENT

SECTION A — FOUR FORCES OF FLIGHT
GL 7 ONLINE — FOUR FORCES OF FLIGHT

❑ Lift
❑ Airfoils
❑ Pilot Control of Lift
❑ Weight
❑ Thrust
❑ Drag
❑ Ground Effect

SECTION B — STABILITY
GL 8 ONLINE — STABILITY

❑ Three Axes of Flight
❑ Longitudinal Stability
❑ Center of Gravity Position
❑ Lateral Stability
❑ Directional Stability
❑ Stalls
❑ Spins
❑ Spin Recovery

SECTION C — AERODYNAMICS OF MANEUVERING FLIGHT
GL 9 ONLINE — AERODYNAMICS OF MANEUVERING FLIGHT

❑ Climbing Flight
❑ Left-Turning Tendencies
❑ Descending Flight
❑ Turning Flight
❑ Load Factor

COMPLETION STANDARDS

- Demonstrate understanding of the four forces of flight, stability, maneuverability, stalls, and spins during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 3A, 3B, and 3C; or online exams for GL 7, 8, and 9. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 4.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 4 — The Flight Environment (Sections A, B, C, D)

Private Pilot Online — Jeppesen Learning Center
Module 4 — The Flight Environment (GL 10, 11, 12, 13)

GROUND LESSON 4

REFERENCES

Private Pilot Textbook/e-Book
Chapter 4 — The Flight Environment (Sections A, B, C, D)

Private Pilot Online — Jeppesen Learning Center
Module 4 — The Flight Environment (GL 10, 11, 12, 13)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Safe and efficient operation of aircraft, including collision avoidance:
  - ◊ Explain collision avoidance procedures, including visual scanning techniques and runway incursion avoidance.
  - ◊ Recall right-of-way rules and minimum safe altitudes.
- Applicable subjects of the Aeronautical Information Manual:
  - ◊ Interpret airport markings, signs, and lighting.
  - ◊ Identify airspace types and operating requirements.
- Aeronautical charts for VFR navigation using pilotage, dead reckoning, and navigation systems:
  - ◊ Interpret aeronautical chart symbology.
  - ◊ Interpret communication and navigation information on aeronautical charts.

CONTENT

SECTION A — SAFETY OF FLIGHT
GL 10 ONLINE — SAFETY OF FLIGHT

❑ Collision Avoidance
❑ Visual Scanning
❑ Cockpit Traffic Displays
❑ Airport Operations
❑ Right-of-Way Rules
❑ Minimum Safe Altitudes
❑ Wire Strike Avoidance
❑ Flight Over Hazardous Terrain
❑ Taxiing in Wind
❑ Positive Exchange of Flight Controls

SECTION B — AIRPORTS
GL 11 ONLINE — AIRPORTS

- ❑ Controlled and Uncontrolled Airports
- ❑ Runway Layout
- ❑ Traffic Pattern
- ❑ Airport Visual Aids
- ❑ Runway and Taxiway Markings
- ❑ Ramp Area Hand Signals
- ❑ Runway Incursion Avoidance
- ❑ Land and Hold Short Operations (LAHSO)
- ❑ Airport Lighting
- ❑ Visual Glideslope Indicators
- ❑ Approach Light Systems
- ❑ Pilot-Controlled Lighting
- ❑ Airport Security

SECTION C — AERONAUTICAL CHARTS
GL 12 ONLINE — AERONAUTICAL CHARTS

- ❑ Latitude and Longitude
- ❑ Projections
- ❑ Sectional Charts
- ❑ VFR Terminal Area Charts
- ❑ Chart Symbology

SECTION D — AIRSPACE
GL 13 ONLINE — AIRSPACE

- ❑ Airspace Classifications
- ❑ Uncontrolled Airspace
- ❑ Controlled Airspace
  - ◊ Class E
  - ◊ Class D
  - ◊ Class C
  - ◊ Class B
  - ◊ Class A
- ❑ Special VFR
- ❑ Special Use Airspace
- ❑ Other Airspace Areas
- ❑ Temporary Flight Restrictions (TFRs)
- ❑ Air Defense Identification Zone (ADIZ)
- ❑ Washington DC Special Flight Rules Area (SFRA)
- ❑ Intercept Procedures

COMPLETION STANDARDS

- Demonstrate understanding of collision avoidance, right-of-way rules, minimum safe altitudes, airport marking and lighting, runway incursion avoidance, LAHSO, aeronautical charts, and airspace requirements during oral quizzing by the instructor.

- Complete with a minimum score of 80 percent: questions for Chapter 4A, 4B, 4C, and 4D; or online exams for GL 10, 11, 12, and 13. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 5.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book

Chapter 5 — Communication and Flight Information (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center

Module 5 — Communication and Flight Information (GL 14, 15, 16)

GROUND LESSON 5

REFERENCES

Private Pilot Textbook/e-Book
Chapter 5 — Communication and Flight Information (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 5 — Communication and Flight Information (GL 14, 15, 16)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Applicable subjects of the Aeronautical Information Manual and the appropriate FAA advisory circulars:
  - ◊ Recognize the characteristics of the ADS-B system and radar operation.
  - ◊ Explain how to properly operate a transponder.
  - ◊ Identify the services provided by ATC and Flight Service.
- Radio communication procedures:
  - ◊ Describe the proper techniques for transmitting on the radio.
  - ◊ Explain the procedures for communicating at controlled and uncontrolled airports.
- Preflight action that includes how to obtain information on runway lengths at airports of intended use and data on takeoff and landing distances:
  - ◊ Recognize the sources of flight information.
  - ◊ Locate flight information by using Chart Supplements, NOTAMs, FARs, the AIM, and advisory circulars.

CONTENT

SECTION A — ATC SERVICES
GL 14 ONLINE — ATC SERVICES

- ❑ ADS-B System
- ❑ Radar
- ❑ Transponders

- ❑ Flight Service
- ❑ Control Tower Services
- ❑ Automatic Terminal Information Service (ATIS)
- ❑ TRACON and ARTCC Services
- ❑ Interpreting Traffic Advisories

SECTION B — RADIO PROCEDURES
GL 15 ONLINE — RADIO PROCEDURES

- ❑ VHF Communication Equipment
- ❑ Using the Radio
- ❑ Phonetic Alphabet
- ❑ Using Numbers on the Radio
- ❑ Coordinated Universal Time (UTC)
- ❑ Common Traffic Advisory Frequency (CTAF)
- ❑ Controlled Airports
- ❑ ATC Facilities
- ❑ Reading Back Clearances
- ❑ Lost Communication Procedures
- ❑ Emergency Procedures and ELTs

SECTION C — SOURCES OF FLIGHT INFORMATION
GL 16 ONLINE — SOURCES OF FLIGHT INFORMATION

- ❑ Locating Flight Information
- ❑ Aeronautical Charts
- ❑ Chart Supplements
- ❑ Airport/Facility Directory
- ❑ Electronic Flight Bag (EFB)
- ❑ Notices to Airmen (NOTAMs)
- ❑ Federal Aviation Regulations (FARs)
- ❑ Aeronautical Information Manual (AIM)
- ❑ Advisory Circulars (ACs)

COMPLETION STANDARDS

- Demonstrate understanding of ATC services, radio procedures and sources of flight information during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 5A, 5B, and 5C; or online exams for GL 14, 15, and 16. With the instructor, review each incorrect response to ensure complete understanding before taking the Stage I Exam in Ground Lesson 6.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book

Review Chapters 1 – 5 in preparation for the Stage I Exam.

Private Pilot Online — Jeppesen Learning Center

Review Modules 1 – 5 (GL 1 – 16) in preparation for the Stage I Exam.

GROUND LESSON 6 — STAGE I EXAM

REFERENCES

Private Pilot Textbook/e-Book
Chapters 1 – 5

Private Pilot Online — Jeppesen Learning Center
Modules 1 – 5 (GL 1 – 16)

OBJECTIVE

Demonstrate knowledge of the subjects covered in Ground Lessons 1 – 5.

CONTENT

If using Private Pilot Online in the Jeppesen Learning Center, you will find the Stage I Exam in Module 6.

STAGE I EXAM

- ❑ Airplane Systems
- ❑ Aerodynamic Principles
- ❑ The Flight Environment
- ❑ Communication and Flight Information

COMPLETION STANDARDS

To complete the lesson and stage, pass the Stage I Exam with a minimum score of 80 percent. With the instructor, review each incorrect response to ensure complete understanding before starting Stage II.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book

Chapter 6 — Meteorology for Pilots (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center

Module 7 — Meteorology for Pilots (GL 17, 18, 19)

Stage II

STAGE OBJECTIVES

During this stage, the student explores weather theory, typical weather patterns, and aviation weather hazards. In addition to meteorological theory, the student learns how to obtain and interpret various weather reports, forecasts, and graphic weather products. Finally, the student becomes thoroughly familiar with the FARs as they apply to private pilot operations.

STAGE COMPLETION STANDARDS

This stage is complete when the student passes the Stage II Exam with a minimum score of 80%, and the instructor has reviewed with the student each incorrect response to ensure complete understanding before starting Stage III.

GROUND LESSON 7

REFERENCES

Private Pilot **Textbook/e-Book**
Chapter **6** — Meteorology for Pilots (Sections A, B, C)

Private Pilot Online — **Jeppesen Learning Center**
Module 7 — Meteorology for Pilots (GL 17, 18, 19)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Recognition of critical weather situations from the ground and in flight and wind shear avoidance:
  - ◊ Describe the causes of various weather conditions, frontal systems, weather patterns, and hazardous weather phenomena.
  - ◊ Explain how to recognize and avoid weather hazards, including thunderstorms, wind shear, turbulence, and restrictions to visibility.
- Safe and efficient operation of aircraft, including recognition and avoidance of wake turbulence:
  - ◊ Recognize how wake is generated.
  - ◊ Identify techniques to avoid wake turbulence.

CONTENT

SECTION A — BASIC WEATHER THEORY
GL 17 ONLINE — BASIC WEATHER THEORY

- ❑ The Atmosphere
- ❑ Atmospheric Circulation
- ❑ Atmospheric Pressure

- ❑ Coriolis Force
- ❑ Global Wind Patterns
- ❑ Local Wind Patterns

SECTION B — WEATHER PATTERNS
GL 18 ONLINE — WEATHER PATTERNS

- ❑ Atmospheric Stability
- ❑ Temperature Inversions
- ❑ Moisture
- ❑ Humidity
- ❑ Dewpoint
- ❑ Clouds and Fog
- ❑ Precipitation
- ❑ Air Masses
- ❑ Fronts

SECTION C — WEATHER HAZARDS
GL 19 ONLINE — WEATHER HAZARDS

- ❑ Thunderstorms
- ❑ Turbulence
- ❑ Wake Turbulence
- ❑ Wind Shear
- ❑ Icing
- ❑ Restrictions to Visibility
- ❑ Volcanic Ash

COMPLETION STANDARDS

- Demonstrate understanding of basic weather theory, weather patterns, and weather hazards during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 6A, 6B, and 6C; or online exams for GL 17, 18, and 19. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 8.

STUDY ASSIGNMENT

FAR/AIM Manual/e-Book

Private Pilot FARs

Private Pilot Online — Jeppesen Learning Center

Module 9 — Federal Aviation Regulations (FARs) (GL 23, 24)

GROUND LESSON 8

REFERENCES

FAR/AIM Manual/e-Book
Private Pilot FARs

Private Pilot Online — Jeppesen Learning Center
Module 9 — Federal Aviation Regulations (FARs) (GL 23, 24)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Applicable Federal Aviation Regulations (FARs) for private pilot privileges, limitations, and flight operations:
  - ◊ Explain the regulations that apply to solo operations and the requirements for private pilot certification and currency in FAR Part 61.
  - ◊ Explain the general operating and flight rules that apply to VFR operations in FAR Part 91.
- Accident reporting requirements of the National Transportation Safety Board (NTSB):
  - ◊ Define the terms used in NTSB 830.
  - ◊ List the incidents that require NTSB notification and the information that must be given in notification.

CONTENT

- ❑ FAR Part 1
- ❑ FAR Part 61
- ❑ FAR Part 91
- ❑ NTSB 830

COMPLETION STANDARDS

- Demonstrate understanding of the relevant regulations in 14 CFR (FAR) Part 1, 61, 91, and 49 CFR (NTSB) 830 during oral quizzing by the instructor.
- If using the Private Pilot online course, complete the online exams for GL 23 and 24 with a minimum score of 80 percent. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 9.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 7 — Interpreting Weather Data (Sections A, B, C, D)

Private Pilot Online — Jeppesen Learning Center
Module 8 — Interpreting Weather Data (GL 20, 21, 22)

GROUND LESSON 9

REFERENCES

Private Pilot Textbook/e-Book
Chapter 7 — Interpreting Weather Data (Sections A, B, C, D)

Private Pilot Online — Jeppesen Learning Center
Module 8 — Interpreting Weather Data (GL 20, 21, 22)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Recognition of critical weather situations from the ground and in flight and the procurement and use of aeronautical weather reports and forecasts:
  - ◊ Explain how to obtain weather reports, forecasts, and graphic products.
  - ◊ Interpret weather reports and forecasts, including recognizing critical weather conditions.
- Preflight action that includes how to obtain weather reports and forecasts:
  - ◊ Identify preflight and inflight weather sources, including Flight Service.
  - ◊ Recognize how to acquire weather data for a specific flight, including obtaining an adequate briefing.

CONTENT

SECTION A — THE FORECASTING PROCESS
GL 20 ONLINE — PRINTED REPORTS AND FORECASTS

- ❑ Forecasting Methods
- ❑ Compiling and Processing Weather Data
- ❑ Forecasting Accuracy and Limitations

SECTION B — PRINTED REPORTS AND FORECASTS
GL 20 — PRINTED WEATHER REPORTS AND FORECASTS

- ❑ Aviation Routine Weather Report (METAR)
- ❑ Radar Weather Reports
- ❑ Pilot Weather Reports (PIREPs)
- ❑ Terminal Aerodrome Forecast (TAF)
- ❑ Winds and Temperatures Aloft Forecast
- ❑ Severe Weather Reports and Forecasts
- ❑ AIRMET/SIGMET/Convective SIGMET

SECTION C — GRAPHIC WEATHER PRODUCTS
GL 21 — GRAPHIC WEATHER REPORTS AND FORECASTS

- ❑ Surface Analysis Chart
- ❑ Weather Depiction Chart
- ❑ Radar Charts and Images
- ❑ Satellite Weather Pictures
- ❑ Significant Weather (SIGWX) Prognostic Charts

- ❑ Convective Outlook Chart
- ❑ National Convective Weather Forecast
- ❑ Forecast Winds and Temperatures Aloft
- ❑ Current and Forecast Icing Products
- ❑ Ceiling and Visibility Analysis
- ❑ Volcanic Ash Forecast and Dispersion Chart

SECTION D — SOURCES OF
GL 22 — SOURCES OF WEATHER INFORMATION

- ❑ Preflight Weather Sources
- ❑ Flight Service
- ❑ In-Flight Weather Sources
- ❑ Automated Weather Reporting Systems
- ❑ Data Link Weather
- ❑ Airborne Weather Radar

COMPLETION STANDARDS

- Demonstrate understanding of the forecasting process, printed reports and forecasts, graphic weather products and sources of weather information during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 7A, 7B, and 7C, and 7D; or online exams for GL 20, 21, and 22. With the instructor, review each incorrect response to ensure complete understanding before taking the Stage II Exam in Ground Lesson 10.

STUDY ASSIGNMENT

Private Pilot and FAR/AIM Books/e-Books

Review Chapters 6 and 7, and the FARs that apply to private pilots in preparation for the Stage II Exam.

Private Pilot Online — Jeppesen Learning Center

Review Modules 7 – 9 (GL 17 – 24) in preparation for the Stage II Exam.

GROUND LESSON 10
STAGE II EXAM

REFERENCES

Private Pilot Textbook/e-Book
Chapters 6 and 7

FAR/AIM Manual/e-Book
Private Pilot FARs

Private Pilot Online — Jeppesen Learning Center
Modules 7 – 9 (GL 17 – 24)

OBJECTIVE
Demonstrate knowledge of the subjects covered in Ground Lessons 7 – 9.

CONTENT
If using Private Pilot Online in the Jeppesen Learning Center, you will find the Stage II Exam in Module 10.

STAGE II EXAM
- ❑ Meteorology for Pilots
- ❑ Federal Aviation Regulations
- ❑ Interpreting Weather Data

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 8 — Airplane Performance (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 11 — Airplane Performance (GL 25, 26, 27)

COMPLETION STANDARDS
This lesson and stage are complete when the student passes the Stage II Exam with a minimum score of 80%, and the instructor has reviewed with the student each incorrect response to ensure complete understanding before starting Stage III.

Stage III

STAGE OBJECTIVES

During this stage, the student learns how to predict performance and control the weight and balance condition of the airplane. In addition, the student explores pilotage, dead reckoning, VOR, and GPS navigation. The student learns how to calculate aircraft weight and balance and performance, as well as plan flights using aeronautical charts, flight computers, and navigation logs. In addition, the student discovers the physiological factors that can affect both the pilot and passengers during flight. Finally, the student learns how to conduct comprehensive preflight planning for cross-country flights, including using SRM tools to manage risk and make effective decision.

STAGE COMPLETION STANDARDS

This stage is complete when the student passes the Stage III Exam with a minimum score of 80%, and the instructor has reviewed with the student each incorrect response to ensure complete understanding before administering the End-of-Course Final Exams.

GROUND LESSON 11

REFERENCES

Private Pilot Textbook/e-Book
Chapter 8 — Airplane Performance (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 11 — Airplane Performance (GL 25, 26, 27)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Preflight action that includes how to obtain data on takeoff and landing distances and fuel requirements / effects of density altitude on takeoff and climb performance:
  - ◊ Use data supplied by the manufacturer to predict airplane performance, including takeoff and landing distances and fuel requirements.
  - ◊ Use a flight computer to calculate airplane performance, such as time, speed, distance, and fuel consumption.
- Weight and balance computations:
  - ◊ Compute and control the weight and balance condition of a typical training airplane.
  - ◊ Recognize the effects of operating at high total weights and with the center of gravity (CG) at positions near the forward and aft limits.

CONTENT

SECTION A — PREDICTING PERFORMANCE
GL 26 ONLINE — PREDICTING PERFORMANCE

- ❑ Aircraft Performance and Design
- ❑ Chart Presentations
- ❑ Factors Affecting Aircraft Performance
- ❑ Takeoff and Landing Performance
- ❑ Climb Performance
- ❑ Cruise Performance

SECTION B — WEIGHT AND BALANCE
GL 25 ONLINE — WEIGHT AND BALANCE

- ❑ Importance of Weight
- ❑ Importance of Balance
- ❑ Weight and Balance Terms
- ❑ Principles of Weight and Balance
- ❑ Determining Total Weight and Center of Gravity
  - ◊ Computation Method
  - ◊ Table Method
  - ◊ Graph Method
  - ◊ Weight-Shift Formula
- ❑ Effects of Operating at High Total Weights
- ❑ Flight at Various CG Positions

SECTION C — FLIGHT COMPUTERS
GL 27 ONLINE — MECHANICAL FLIGHT COMPUTERS

- ❑ Mechanical Flight Computers
- ❑ Time, Speed, and Distance
- ❑ Fuel Consumption
- ❑ Airspeed and Density Altitude Computations
- ❑ Wind Problems
- ❑ Conversions
- ❑ Electronic Flight Computers

COMPLETION STANDARDS

- Calculate weight and balance and determine airplane performance under a variety of conditions and explain the results during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 8A, 8B, and 8C; or online exams for GL 25, 26, and 27. With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 12.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 9 — Navigation (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 12 — Navigation (GL 28, 29, 30, 31)

GROUND LESSON 12

REFERENCES

Private Pilot Textbook/e-Book
Chapter 9 — Navigation (Sections A, B, C)

Private Pilot Online — Jeppesen Learning Center
Module 12 — Navigation (GL 28, 29, 30, 31)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Applicable Federal Aviation Regulations for flight operations:
  - ◊ Plan a VFR flight using pilotage, dead reckoning, and navigation aids, considering fuel reserves and the appropriate VFR cruising altitude.
  - ◊ Create a nav log and fill out a flight plan form.
- Applicable subjects of the Aeronautical Information Manual and the appropriate FAA advisory circulars:
  - ◊ Explain the operation of navigation aids—VOR and GPS.
  - ◊ Interpret navigation aids—VOR and GPS—for VFR navigation.
- How to plan for alternatives if the planned flight cannot be completed or delays are encountered.
  - ◊ Identify the steps to perform lost procedures.
  - ◊ Recognize how to monitor your flight progress and the actions to take if your flight is delayed or cannot be completed as planned.

CONTENT

SECTION A — PILOTAGE AND DEAD RECKONING
GL 28 ONLINE — PILOTAGE AND DEAD RECKONING

- ❑ Pilotage
- ❑ Dead Reckoning
- ❑ VFR Cruising Altitudes
- ❑ Flight Planning
- ❑ VFR Flight Plan
- ❑ Lost Procedures

SECTION B — VOR NAVIGATION
GL 29 ONLINE — VOR NAVIGATION
GL 31 ONLINE — ADF NAVIGATION

- ❑ Ground Equipment
- ❑ Airborne Equipment
- ❑ Navigation Procedures
- ❑ Checking VOR Accuracy
- ❑ Horizontal Situation Indicator (HSI)
- ❑ Distance Measuring Equipment (DME)
- ❑ ADF Navigation

ADF navigation content is optional based on the instructor's discretion, flight environment, and aircraft equipment.

SECTION C — SATELLITE NAVIGATION — GPS
GL 30 ONLINE — GPS NAVIGATION

- ❑ GPS Operation
- ❑ WAAS
- ❑ RAIM
- ❑ Navigating with GPS
- ❑ Navigation Database
- ❑ Course Deviation Indicator (CDI)
- ❑ Moving Map
- ❑ Waypoints
- ❑ GPS Flight Planning
- ❑ Navigation Data
- ❑ Intercepting and Tracking a Course

COMPLETION STANDARDS

- Complete a nav log and flight plan as assigned by the instructor and demonstrate understanding of pilotage and dead reckoning during oral quizzing by the instructor.
- Demonstrate understanding of VOR navigation, GPS navigation, and ADF navigation (if applicable), during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 9A, 9B, and 9C; or online exams for GL 28, 29, 30 (the online course does not test on ADF navigation). With the instructor, review each incorrect response to ensure complete understanding before starting Ground Lesson 13.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 10 — Applying Human Factors Principles (Sections A, B)

Private Pilot Online — Jeppesen Learning Center
Module 13 — Applying Human Factors Principles (GL 32, 33)

GROUND LESSON 13

REFERENCES

Private Pilot Textbook/e-Book
Chapter 10 — Applying Human Factors Principles (Sections A, B)

Private Pilot Online — Jeppesen Learning Center
Module 13 — Applying Human Factors Principles (GL 32, 33)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Applicable subjects of the Aeronautical Information Manual and the appropriate FAA advisory circulars:
  - ◊ Recognize how the body functions in flight, including the limitations of vision, disorientation, and respiration.
  - ◊ Identify the causes, symptoms, and treatments for hypoxia and hyperventilation.
- Aeronautical decision making and judgment:
  - ◊ Explain how to use the tools and techniques of SRM to perform these tasks: risk management, task management, maintain situational awareness, CFIT awareness, and automation management.
  - ◊ Apply the aeronautical decision making process to make effective choices during flight operations.

CONTENT

SECTION A — AVIATION PHYSIOLOGY

- ❑ Vision in Flight
- ❑ Night Vision
- ❑ Visual Illusions
- ❑ Disorientation
- ❑ Respiration
- ❑ Hypoxia
- ❑ Hyperventilation

SECTION B — SINGLE-PILOT RESOURCE MANAGEMENT (SRM)

- ❑ Accidents and Incidents
- ❑ Aeronautical Decision Making (ADM)
- ❑ Self-Assessment
- ❑ Hazardous Attitudes
- ❑ Risk Management
- ❑ Task Management
- ❑ Situational Awareness
- ❑ CFIT Awareness
- ❑ Automation Management
- ❑ Applying SRM

COMPLETION STANDARDS

- Demonstrate understanding of aviation physiology and SRM tools and techniques during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 10A and 10B; or online exams for GL 32 and 33. Review with the instructor each incorrect response to ensure complete understanding before starting Ground Lesson 14.

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book
Chapter 11 — Flying Cross-Country (Sections A, B)

Private Pilot Online — Jeppesen Learning Center
Module 14 — Flying Cross-Country (GL 34, 35)

GROUND LESSON 14

REFERENCES

Private Pilot Textbook/e-Book
Chapter 11 — Flying Cross-Country (Sections A, B)

Private Pilot Online — Jeppesen Learning Center
Module 14 — Flying Cross-Country (GL 34, 35)

OBJECTIVES

Part 141/61 Aeronautical Knowledge

- Preflight actions that include (1) how to obtain information on runway lengths at airports of intended use, data on takeoff and landing distances, weather reports and forecasts, and fuel requirements; and (2) how to plan for alternatives if the planned flight cannot be completed or delays are encountered:
  - ◊ Gain proficiency in planning a cross-country flight.
  - ◊ Identify the tasks that apply to flying a typical cross-country flight, including locating checkpoints, making in-flight time and fuel calculations, and evaluating weather conditions.
  - ◊ Recognize how to make and implement decisions regarding alternative actions during a cross-country flight, such as implementing a diversion.
- Applicable subjects of the Aeronautical Information Manual and the appropriate FAA advisory circulars; applicable Federal Aviation Regulations for private pilot privileges, limitations, and flight operations:
  - ◊ Identify the pilot responsibilities that apply to planning and performing a cross-country flight.
  - ◊ Apply knowledge of airport operations, airspace, ATC services, and flight information sources to plan and perform a cross-country flight.

CONTENT

SECTION A — THE FLIGHT PLANNING PROCESS
GL 34 ONLINE— THE FLIGHT PLANNING PROCESS

- ❑ Perform a Flight Overview
- ❑ Develop the Route
- ❑ Obtain a Weather Briefing
- ❑ Complete the Nav Log
- ❑ File the Flight Plan
- ❑ Perform Preflight Tasks

SECTION B — THE FLIGHT
GL 35 ONLINE — THE FLIGHT

- ❑ Predeparture
- ❑ Climb and Initial Cruise
- ❑ Enroute
- ❑ Diversion
- ❑ Descent
- ❑ Before Approach and Landing
- ❑ Postflight

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book

Review Chapters 8 – 11 in preparation for the Stage III Exam.

Private Pilot Online — Jeppesen Learning Center

Review Modules 11 – 14 in preparation for the Stage III Exam.

COMPLETION STANDARDS

- Demonstrate understanding of the flight planning process and of flying a cross-country during oral quizzing by the instructor.
- Complete with a minimum score of 80 percent: questions for Chapter 11A and 11B; or online exams for GL 34 and 35. Review with the instructor each incorrect response to ensure complete understanding before taking the Stage III Exam in Ground Lesson 15.

GROUND LESSON 15

STAGE III EXAM

REFERENCES

Private Pilot Textbook/e-Book
Chapters 8 – 11

Private Pilot Online — Jeppesen Learning Center
Modules 11 – 14 (GL 25 – 35)

OBJECTIVE

Demonstrate knowledge of the subjects covered in Ground Lessons 11 – 14.

CONTENT

If using Private Pilot Online in the Jeppesen Learning Center, you will find the Stage III Exam in Module 15.

STAGE III EXAM

- ❑ Airplane Performance

- ❑ Navigation
- ❑ Applying Human Factors Principles
- ❑ Flying Cross-Country

STUDY ASSIGNMENT

Private Pilot Textbook/e-Book

Private Pilot Online — Jeppesen Learning Center

Review the entire textbook or online course, as necessary, in preparation for the End-of-Course Final Exam "A."

COMPLETION STANDARDS

This stage is complete when the student passes the Stage III Exam with a minimum score of 80%, and the instructor has reviewed with the student each incorrect response to ensure complete understanding before administering the End-of-Course Final Exams.

GROUND LESSON 16

END-OF-COURSE FINAL EXAM "A"

REFERENCES

Private Pilot Textbook/e-Book
Chapters 1 – 11

Private Pilot Online — Jeppesen Learning Center
Modules 1 – 14 (GL 1 – 35)

OBJECTIVE

Demonstrate comprehension of the material presented in this course in preparation for the FAA Private Pilot Airman Knowledge Test.

CONTENT

- ❑ Private Pilot End-of-Course Final Exam "A"

STUDY ASSIGNMENT

Review any deficient subject areas based on the results of End-of-Course Final Exam "A." Review in preparation for End-of-Course Final Exam "B."

COMPLETION STANDARDS

Complete the End-of-Course Final Exam "A" with a minimum score of 80% and review with the instructor each incorrect response to ensure complete understanding before taking the End-of-Course Final Exam "B."

GROUND LESSON 17
END-OF-COURSE FINAL EXAM "B"

REFERENCES

Private Pilot Textbook/e-Book
Chapters 1 – 11

Private Pilot Online — Jeppesen Learning Center
Modules 1 – 14 (GL 1 – 35)

OBJECTIVES
Demonstrate comprehension of the material presented in this course in preparation for the FAA Private Pilot Airman Knowledge Test.

CONTENT
❑ Private Pilot End-of-Course Final Exam "B"

STUDY ASSIGNMENT
Review any deficient subject areas based on the results of End-of-Course Final Exam "B." Review in preparation for the FAA Private Pilot Airman Knowledge Test.

COMPLETION STANDARDS
Complete the End-of-Course Final Exam "B" with a minimum score of 80% and review with the instructor each incorrect response to ensure complete understanding so that the instructor can provide recommendation to take the FAA Private Pilot Airman Knowledge Test.